Details

1. Manufacturing Strategy and Material Selection

- Use medical-grade resins (ISO 10993-5/-10 or USP Class VI) such as PC-ISO or Loctite MED412 for the structural shell.
- TPU overmolding should be optically clear and IR-permissive (e.g., Dow Silastic, 0.3 mm thick).
- Maintain total enclosure thickness under 2.5 mm by controlling resin wall thickness (~0.4 mm) and using thin adhesive layers.
- Conduct bonding trials between TPU and resin to ensure watertight seals and thermal resistance under use conditions.
- Final sealing methods should prioritize ultrasonic welding or UV epoxy curing.

2. Mechanical Design and Assembly

- The full assembly must fit within a 24 mm diameter circle and 2.5 mm stack height.
- Position the MAX86178 sensor in direct alignment under the optical TPU window.
- Design internal resin ridges or light-blocking channels around the sensor cavity to reduce internal reflection.
- Battery placement should be centrally located and thermally isolated.
- Keep the MDBT42V module centered for signal path balance and RF symmetry.
- Use a keyed or slotted mount for PCB-to-housing alignment and final bonding.
- Include adhesive fixture or mounting bosses in the resin shell for better handling.

3. Sensor Layout and Antenna Optimization

- Place the MAX86178 near the base (skin-facing side), ensuring an unobstructed optical path.
- Keep the LIS2DW12 IMU offset from BLE antenna traces to prevent EMI.
- The MDBT42V BLE module should be placed near the top edge with at least 5 mm RF clearance to ground planes and batteries.
- Avoid copper pour beneath or adjacent to its antenna.

- Optional u.FL pad may be included as fallback.

- Position decoupling capacitors near MAX86178 and PMIC, with GND stitching via arrays to stabilize analog lines.

4. Electrical Engineering Implementation

- Use a one-sided 0.6 mm FR4 board with components all facing the resin cavity side.

- Follow MDBT42V antenna keep-out guidelines strictly.

- Power the board using a soft-pack LiPo battery (0.9-1.0 mm thick) routed through the nPM1100 PMIC.

- Route I2C bus for MAX86178 and LIS2DW12.

- Add external pull-up resistors and EMI protection capacitors to both I2C lines.

- Include in-circuit SWD debug pads positioned within the board outline at ~45 orientation for fixture access.

- Add filtering caps to MAX86178 LED power line and all MCU VDD/GND junctions.

5. Firmware Architecture and Power Strategy

- Use the Nordic nRF5 SDK or Zephyr for firmware development.

- Configure MAX86178 in green LED-only mode at 25-50 Hz sampling frequency.

- Use LIS2DW12 to wake the MCU on motion interrupts and classify activity context.

- Advertise BLE with intervals of 500 ms during idle and 100 ms when active.

- Implement Nordic Secure DFU bootloader with firmware rollback logic.

- Add watchdog timers and BLE fail-recovery logic to reboot gracefully on signal loss or firmware error.

- Target average current < 300 A with deep sleep idle below 5 A.

6. Mobile App Development Requirements

- Design a cross-platform BLE app (Swift/Kotlin/Flutter).

- Scan for devices broadcasting as "NestorHR-XXXX".

- Display heart rate (HR), battery % (via 0x2A19), and firmware version.

- Implement automatic reconnect logic and use the Nordic DFU library for seamless OTA updates.

- BLE UUIDs should be readable immediately upon connect.

- Include BLE activity logs in the app for user-side troubleshooting.

- Ensure iOS/Android permissions are declared correctly (location, background BLE) to maintain App Store compliance.

7. Certification and Regulatory Path

- To avoid full retesting, retain MDBT42V's radio and antenna configuration exactly.

- Register your product with the Bluetooth SIG using Declaration Only, referencing QDID 89121.

- Use Raytac's FCC ID (SH6MDBT42V) and CE/MIC/IC filings.

- Print compliant labels and include FCC and CE notices in your documentation.

- If BLE antenna or power output is changed, prepare for RF re-certification.

- RoHS and REACH documentation should be maintained for material compliance.

8. Final Assembly, Testing, and QA Strategy

- Assemble with TPU overmold first, then install the PCB and battery.

- Use optical alignment fixtures to align PPG sensor directly under window.

- Use thermal bonding or foam compression for battery retention.

- Run BLE range test, battery charging, HR detection, and IMU interrupts in end-of-line (EOL) test fixture.

- Include visual inspection for resin fill, optical clarity, and pad corrosion.

- Seal using laser welding or UV epoxy with jigs to maintain constant pressure and geometry during cure.